Edge Computing

Limitations of Cloud-Only Architectures

- **High latency:** Centralized data centers introduce delays unsuitable for time-critical applications
- **Network dependency:** Full connectivity required; outages can halt operations
- **Bandwidth constraints:** Transmitting large volumes of data to the cloud is costly and inefficient
- **Privacy concerns:** Sensitive data may traverse multiple jurisdictions
- **Scalability limitations:** Centralized cloud may struggle with real-time, high-frequency workloads

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Definition of Edge Computing

Edge computing is a distributed computing paradigm in which computational tasks and data storage are performed closer to the sources of data, rather than solely in centralized cloud servers.

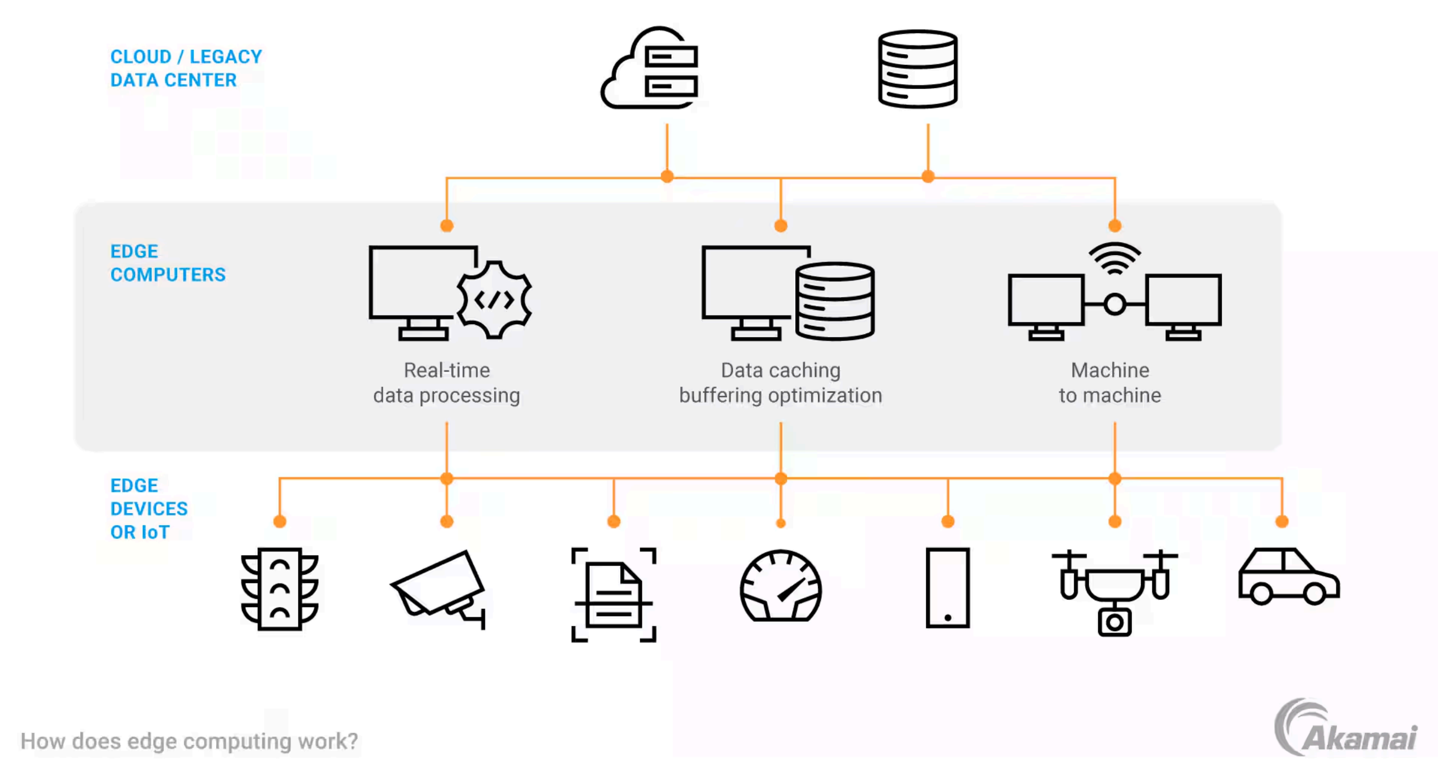
Deployment locations include:

- Industrial controllers and IoT devices
- Embedded systems and on-premise servers
- Micro data centers and regional/5G edge nodes

Edge-Cloud Complementarity

- **Edge:** Executes **time-critical**, **local**, **bandwidth-sensitive**, and **privacy-sensitive** operations
- **Cloud:** Performs **global coordination**, **long-term storage**, **machine learning training**, and **orchestration**

Key Benefit: This hybrid model reduces latency, conserves network bandwidth, enhances reliability, and preserves data privacy, while retaining the computational power and orchestration capabilities of the cloud.



Edge Computing is Inherently Distributed

Edge computing is a form of **distributed system**: instead of one centralized system, computation is spread across many geographically dispersed nodes. This naturally introduces classic distributed systems challenges:

Multiple Nodes Working Together

- remote communication
- state synchronization
- coordination mechanisms

Heterogeneity

- small embedded devices
- powerful on-prem servers
- cloud services

Intermittent Connectivity

- go offline
- operate in degraded network conditions
- sync periodically